

Capstone Data Dictionary

Every column, and when its value is recorded

Read this before you model. The second column of every table below - when the value is recorded - matters more than the description. It is what tells you whether a column can honestly be used to predict a given target.

THE DAY-BEFORE TEST For any column you are thinking of feeding a model, ask: would this value exist, with this value, the day before the thing I am predicting happened? If the answer is no, the column is a consequence of the outcome, not a predictor of it, and using it will give you an excellent score and a worthless model.

1. capstone_subscribers.csv

One row per subscriber. 3222 rows in the file, 3200 distinct subscribers - the difference is 22 exact duplicate rows introduced by a double-write in the extract. Behavioural columns summarise the six months to 31 July 2026.

Column	Recorded when	Description
customer_id	At activation	Unique subscriber key. Not unique in the raw file - de-duplicate first.
home_cell_id	Continuously	The cell serving the subscriber's registered address. Joins to the network files.
region	At activation	Circle. Present in several spellings and cases; standardise before grouping.
plan_type	At activation, on change	Prepaid or Postpaid.
device_type	On handset change	4G Feature, 4G Smart or 5G Smart.
contract_type	At activation, on change	Monthly or Annual. Prepaid subscribers are always Monthly.
tenure_months	Continuously	Months since activation, 1 to 72.
arpu	Monthly billing run	Average revenue per user per month, in rupees. Target for a regression task.
avg_monthly_gb	Monthly	Mean data used per month. 99 values are blank.
avg_voice_min	Monthly	Mean outgoing voice minutes per month.
avg_sms	Monthly	Mean SMS sent per month.
roaming_days_6m	Monthly	Days on roaming in the last six months.
complaints_6m	On each complaint	Complaints raised in the last six months, any category.
network_tickets_6m	On each ticket	Complaints specifically about network quality.
payment_failures_6m	On each failure	Failed payment attempts in the last six months.

Column	Recorded when	Description
data_pack_purchases_6m	On each purchase	Add-on data packs bought in the last six months.
days_since_last_recharge	Continuously	Days since the last recharge or payment. 58 values are blank.
autopay_flag	On enrolment	1 if the subscriber has standing payment instructions.
retention_offer_sent	See warning below	1 if a retention offer was issued during the window.
total_charges	Monthly billing run	Cumulative billed amount since activation.
churn	End of window	1 if the subscriber left during the window. The target for classification.

TWO COLUMNS FAIL THE DAY-BEFORE TEST retention_offer_sent is issued by the retention team after somebody has already decided a subscriber is at risk. It went to 84 per cent of subscribers who churned and 5 per cent of those who stayed. It is a record of the operator's own reaction, not of the subscriber's behaviour, and on the day you actually need a churn score it has not been sent yet.

total_charges is arpu multiplied by tenure_months. It is perfectly valid for revenue reporting and completely disqualified as a feature when the target is arpu - the implied ARPU correlates with the true value at 0.9995.

Leakage is a property of the pairing of feature and target, not of the column on its own. Re-run your audit whenever the target changes.

2. capstone_usage_monthly.csv

One row per subscriber per month, six months from February to July 2026. 19238 rows in the file, 19200 after removing 38 duplicate (customer_id, month) pairs.

Column	Recorded when	Description
customer_id	-	Joins to the subscriber file.
month	-	Billing month, YYYY-MM.
data_gb	Month end	Data consumed in the month.
voice_min	Month end	Domestic outgoing voice minutes.
intl_min	Month end	International outgoing minutes.
sms_count	Month end	SMS sent.
revenue_inr	Billing run	Amount billed for the month. 71 values are blank; a further set are exactly zero, which is a different thing entirely - see below.
failed_payment_count	On each failure	Failed payment attempts in the month.
days_active	Month end	Days with any recorded activity.

BLANK AND ZERO ARE NOT THE SAME A blank revenue value means the billing extract did not return a figure - a collection fault. A revenue value of exactly zero on a month with real usage means the subscriber consumed service and was not billed for it. The first is missing data. The second is money the operator lost, and finding it is part of the exercise. 77 rows carry an exact zero.

3. capstone_network_kpi.csv

One row per cell per fifteen-minute interval, across 18 cells and five days (1 to 5 August 2026). 8681 rows in the file, 8640 after removing 41 duplicate (cell_id, reading_ts) pairs left by a double-writing collector.

Column	Recorded when	Description
reading_ts	Every 15 minutes	Interval start, YYYY-MM-DD HH:MM:SS.
cell_id	-	Cell identifier. Joins to home_cell_id in the subscriber file.
region	-	Circle the cell sits in.
latency_ms	Every 15 minutes	Mean round-trip latency. Blank in 90 de-duplicated rows - for three different reasons.
throughput_mbps	Every 15 minutes	Mean cell throughput. 9 rows are negative, which is a counter rollover rather than a measurement.
downtime_min	Every 15 minutes	Minutes of the interval the cell was off air. Zero for a healthy cell.
drop_call_rate_pct	Every 15 minutes	Percentage of calls dropped in the interval.
active_users	Every 15 minutes	Approximate concurrent users on the cell.

A BLANK LATENCY HAS THREE POSSIBLE MEANINGS 20 blanks fall inside a planned maintenance window. The cell was switched off by an engineer holding a change ticket. There was no traffic, so there was no latency to measure, and NULL is the honest record. Imputing these invents a measurement of something that was not happening. 7 blanks fall during an unplanned outage - downtime_min is above zero and no change ticket exists. These are the outage itself, and they are the most important rows in the file. 63 blanks have no explanation at all. These are genuine collector losses and the only ones worth raising a ticket against. Counting these is order-dependent: take the counts before de-duplication and you get different figures, because some blanks sit on duplicated rows. Neither answer is wrong, but only one answers the question you were asked. State the grain, state the operation, then count.

4. capstone_maintenance_windows.csv

Three planned outages, one row each. In production this is not a CSV - it is a live feed from the change management system, and the suppression rule reads it continuously. That integration is the single highest-value piece of alerting engineering most operators never build.

Column	Description
cell_id	The cell taken out of service.
window_start	Start of the planned outage, inclusive.
window_end	End of the planned outage, exclusive.
reason	Free text describing the work.
change_ticket	The approved change reference. A cell down without one of these is an incident.

5. capstone_cell_traffic_monthly.csv

One row per cell per month, thirty months to July 2026. 540 rows, of which 24 have a blank Traffic_GB - interpolate rather than drop, or you will break the even monthly spacing every forecasting method depends on.

Column	Recorded when	Description
Month	Month end	YYYY-MM.
Cell	-	Cell identifier, matching the KPI file.
Region	-	Circle.
Traffic_GB	Month end	Total traffic carried by the cell in the month.
Capacity_GB	On engineering change	Engineered monthly capacity. Constant per cell across the file.

6. capstone_messages.csv

354 free-text customer messages received over five days, with the true sentiment attached so you can measure your own reader against it.

Column	Description
message_id	Unique message key.
customer_id	Joins to the subscriber file. 18 messages carry an id with no matching subscriber.
message_text	What the customer wrote.
true_sentiment	Negative, Positive or Neutral, assigned when the message was composed.
case_type	standard or hard. The 24 hard cases contain negation, litotes or sarcasm, and exist to break a word-counting model. Report accuracy separately for the two groups.
received_at	Timestamp of receipt.

7. capstone_tickets.csv

548 support tickets across 5 handling teams, labelled with the team that actually dealt with each one.

Column	Description
ticket_id	Unique ticket key.

Column	Description
customer_id	Joins to the subscriber file.
ticket_text	What the customer wrote.
team	The team that handled it: Billing, Network, SIM_Activation, Plan_Change or Retention.
created_at	Timestamp of receipt.
case_type	standard, noisy, ambiguous or reassigned - see the note below.

WHY YOU CANNOT REACH 100 PER CENT, AND SHOULD NOT TRY 36 tickets are ambiguous: they name two topics and a human would hesitate over them too.
56 are noisy: they belong clearly to one team but borrow another team's vocabulary.
56 were reassigned: a human router sent them to a different team because of an open case, an account history or a supervisor instruction that the message text never carried.
No model can recover information that is not in the text. The ceiling is real, and it is the reason a confidence floor and a human review path exist. Treat a low-confidence prediction as a design feature, not a failure.